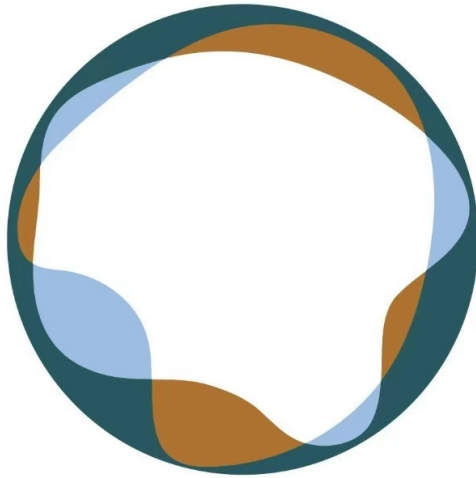




Moonshot Academy

**Exploring the Patterns between
NDVI and LST of Eight Cities in China
Using Multivariable Calculus**

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Courses:

ENVS5001-Advanced Environmental Science

MATH6003-Advanced Multivariable Calculus

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Abstract

abstract

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1 Introduction

Urban Heat Islands (UHIs) refer to metropolitan areas that experience significantly higher temperatures than their rural surroundings, primarily due to human activities and urban infrastructure. This phenomenon has drawn increasing attention due to its impact on public health, energy consumption, and urban sustainability. Among the many variables contributing to UHI formation, vegetation coverage and land surface temperature (LST) are particularly prominent and measurable.

The Normalized Difference Vegetation Index (NDVI) and Land Surface Temperature (LST) are two critical indicators derived from satellite observations. NDVI, defined as $(\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$, reflects the density and vitality of green vegetation. LST represents the radiative skin temperature of the land surface and is commonly used to monitor urban thermal conditions. Both indicators are provided by NASA's MODIS (Moderate Resolution Imaging Spectroradiometer) sensors and are widely utilized in urban climate studies.

According to Yang et al. [1], a recent study published in *Nature*, the spatial and temporal evolution patterns of UHIs were examined across seasons. Their analysis of a major Chinese megacity employed factor and interaction analysis to investigate the relationship between NDVI and LST. The findings demonstrated a statistically significant negative correlation, with a Pearson coefficient of -0.709, reinforcing the understanding that vegetation mitigates urban warming by contributing to surface cooling. Another study from *Nature* introduced a novel Comprehensive Urbanization-Vegetation Coordination Index (CUVCI), combining the Comprehensive Urbanization Level (CUL) with the Compound Annual Growth Rate (CAGR) to capture both spatial and temporal dynamics of urbanization and vegetation. By applying CUVCI alongside NDVI data across the Yellow River Basin from 2000 to 2019, Wang, H., Yang, C., Sun, Y. *et al.* conducted spatiotemporal and statistical analyses to evaluate coordination levels and identify key driving factors.[2] Moreover, a study, which also focuses on the relationship between vegetation and surface temperature, highlights that urban vegetation typically forms mixed, heterogeneous patterns rather than uniform patches, and that understanding how different vegetation configurations influence urban temperatures is essential for planning. Recognizing that linear regression may not effectively capture the complex, nonlinear interactions between vegetation mosaics and surface temperature, the authors applied Random Forest analysis, an ensemble-learning model, to summer daytime Land Surface Temperature and vegetation data from the 2007, 2009, and 2016 Urban Monitor datasets.[3]

However, despite growing interest in the relationship between urban vegetation and surface temperature, existing approaches remain constrained in several important ways. For instance, some studies rely primarily on linear regression, which assumes a straightforward, two-dimensional association between NDVI and LST. This simplification neglects the inherently nonlinear and multidimensional nature of urban environmental systems. Moreover, many analyses fail to explicitly incorporate built-up surfaces as independent variables, despite their significant role in determining surface thermal charac-

teristics. In reality, urban heat dynamics are driven by complex, interacting factors-including vegetation, impervious surfaces, and spatial configuration-necessitating the use of more advanced modeling techniques capable of capturing these intricate relationships. Additionally, conventional mathematical modeling often depends on predefined functional forms and assumptions (e.g., linearity, independence), which may not adequately reflect real-world variability. Therefore, the objective of this study is to construct a multivariate nonlinear model that simultaneously assesses the influence of NDVI on LST across eight cities. By analyzing the coefficients of the fitted models in each city using Principal Components Analysis (PCA), this paper seeks to evaluate whether the relationship between vegetation index and surface temperature demonstrates location-specific variability or a consistent structural pattern.

By restricting the study to large cities located within a relatively narrow geographic corridor (110°E-122°E longitude, 30°N-32°N latitude), with low average altitude, plain landforms, and comparable city size and population, this research hypothesizes that the nonlinear coefficients linking NDVI and LST will demonstrate consistent trends across cities. If supported, these patterns would suggest that, under similar topographic and climatic conditions, vegetation indices can serve as reliable and generalizable predictors of urban thermal behavior.

In contrast to existing research that relies primarily on linear regression or visual correlation, this study employs high-resolution satellite data from NASA's MODIS products and rigorous mathematical modeling to derive a deeper, quantifiable understanding of urban heat mechanisms. The significance of this work lies in its potential to inform data-driven greening strategies and zoning policies by identifying spatially differentiated drivers of urban thermal anomalies.

2 Methodology

To investigate the spatial and temporal dynamics of urban heat islands (UHIs) across multiple cities, this study relies on multi-source remote sensing and statistical data. Ensuring data compatibility, reliability, and temporal coverage is essential for building consistent cross-city comparisons. This section outlines the specific data sources used, the rationale for selecting certain satellite products over others, and the platforms employed for data access and processing.

2.1 Data Source and Selection

Given the global scope of the datasets utilized in this study, the selection of data sources was guided by the need for accuracy, consistency, and temporal coverage. The primary environmental variables-Normalized Difference Vegetation Index (NDVI) and Land Surface Temperature (LST)-were both derived from NASA's MODIS (Moderate Resolution Imaging Spectroradiometer) Terra

satellite products, which are well-established for global-scale ecological and climatic research due to their long temporal span and robust data quality.

Although higher-resolution alternatives such as Landsat 8 exist, MODIS products were preferred in this study for several reasons. First, MODIS offers superior temporal resolution, providing continuous monthly composites, which are essential for detecting seasonal trends and long-term urban heat island (UHI) dynamics. In contrast, Landsat’s 16-day revisit cycle and cloud interference limitations can lead to gaps in temporal continuity, especially in subtropical and monsoon climates. Secondly, while MODIS does offer coarser spatial resolution, this was not a critical limitation for city-level analysis, where regional trends take precedence over localized pixel-level detail.

Among MODIS datasets, MOD13C2 and MOD11C3 were specifically selected. MOD13C2 provides global monthly NDVI composites, and MOD11C3 delivers corresponding monthly LST data. These products share the same spatial resolution (0.05°) and temporal scale (monthly), ensuring compatibility for integrative modeling and comparative analysis. Their consistency in temporal coverage and global reach makes them particularly suitable for analyzing multiple cities across similar latitudinal zones.

All satellite data processing and extraction were conducted via the Google Earth Engine (GEE) platform [4], which offers cloud-based access to petabyte-scale environmental datasets. National-level land and population statistics were retrieved from the National Bureau of Statistics of China (NBS) [5], providing demographic context and allowing for the incorporation of urban-scale variables into the modeling framework.

Variable	Database	Provider	Data Type		Function
NDVI	MOD13C2 v061	GEE[4]	Terra Monthly 0.05 Deg	NDVI L3 Global CMG	Reflects regional vegetation coverage; values range from -1 to 1.
LST	MOD11C3 v061	GEE[4]	Terra Monthly Global 0.05 Deg CMG	LST/Emissivity L3 CMG	Represents urban surface temperature; unit in Kelvin (K).

Table 1: Summary of Data Sources

2.1.1 Study Area

Although the MODIS NDVI and LST datasets provide global coverage, the selection of study areas was constrained by climatic, geographic, and topographic variability, all of which can influence LST. To reduce confounding variables and maintain regional consistency, the geographic focus was limited to China, a country characterized by a pronounced monsoonal climate and diverse urbanization patterns.

Factors	Criteria in scale
Longitude	110°E - 122°E
Latitude	30°N - 32°N
Average altitude	< 30 meters
Landform	plain
City population	large city (population > 1 million)
City size	4000m ² - 9000m ²

Table 2: Criteria of city selection

Initial efforts to select representative cities began by focusing on those located along the same latitude as Beijing, with the goal of controlling for solar radiation and climatic variability. However, this approach soon proved problematic. Northern China features complex topography, especially with widespread mountainous regions, which introduce significant elevation differences that can confound surface temperature comparisons. Moreover, within similar latitudinal and longitudinal ranges in the north, it was difficult to identify other cities that could match Beijing in terms of population size. The stark contrasts in urban development and prosperity levels further limited the comparability of potential candidates. As a result, cities in northern China were excluded to maintain greater consistency in topographic, climatic, and socio-economic conditions.

Based on instructor guidance and preliminary data evaluations, the geographic focus shifted to southern and southeastern China, where cities are more densely distributed and situated on relatively uniform terrain. Shanghai was selected as the anchor city due to its extensive urban development, population density, and availability of high-resolution environmental data.

To establish a consistent spatial framework, two latitude-based filters were applied. First, to define the city center region, a latitude range of 30.5°N to 31.5°N was established. Second, for broader urban extent, a total urban land area range of 30°N to 32°N was used. In parallel, to avoid the effects of China's complex altitudinal gradients, the study was geographically limited to the third step of

China's terrain classification—specifically, areas east of 110°E longitude. This ensured that candidate cities remained within an alluvial plain, a relatively low-elevation, topographically similar zone.

Applying these criteria yielded an initial pool of 16 cities. To further refine the selection and improve geographic comparability, additional filtering was conducted based on average elevation. Cities with significant mountainous terrain or abrupt elevation changes were excluded. Preference was given to cities located on plains, defined here as having an average elevation below 30 meters. This final screening resulted in a carefully selected sample of eight cities: Shanghai, Suzhou, Jiaxing, Huzhou, Wuhu, Wuxi, Wuhan, and Xiaogan. These cities are all located east of 110°E, with city centers between 30.5°N and 31.5°N, and primarily flat topography.

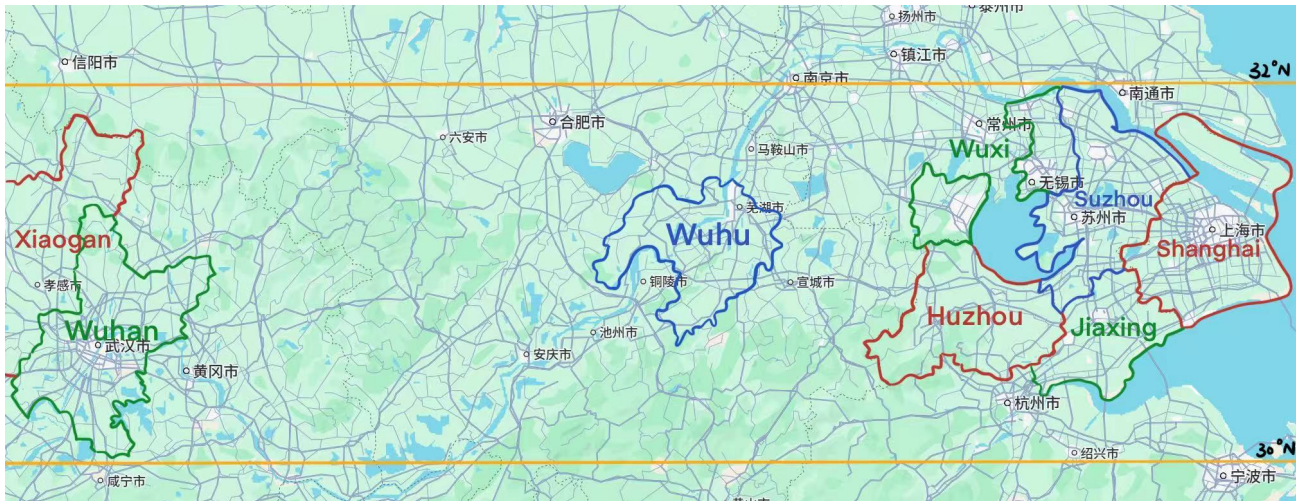


Figure 1: Overview of eight selected cities

Moreover, the selected cities provide a diverse yet coherent representation of China's urban typology. According to the "Notice on Adjusting the Standards for Classifying Urban Sizes" by the National Bureau of Statistics of China, urban areas are categorized based on the number of permanent urban residents (PUR). Within the selected cities:

Megacities (PUR > 10 million): Shanghai and Wuhan

Type I Large Cities (PUR between 5 and 10 million): Suzhou, Wuxi

Type II Large Cities (PUR between 3 and 5 million): Jiaxing

Type III Large Cities (PUR between 1 and 3 million): Wuhu, Xiaogan, Huzhou

This classification ensures that the selected sample not only controls for geographic and climatic variation but also captures meaningful differences in urban scale and population density. Additionally, land area varies across the eight cities—from 4,237 to 8,904 km^2 —providing further variables for evaluating how spatial extent may influence urban heat island dynamics.

City	Province	Lat (°N)	Lon (°E)	avg. Alt (m)	LA (km ²)	PD (people/km ²)
Huzhou	Zhejiang	30.8985	120.0583	3.00	5820.00	1015.27
Jiaxing	Zhejiang	30.7544	120.7494	3.70	4237.00	3110.48
Shanghai	Shanghai	31.2561	121.4709	4.00	6340.50	3923.11
Suzhou	Jiangsu	31.2834	120.3219	4.00	8657.32	1728.91
Wuhan	Hubei	30.5847	114.4346	20.00	8569.19	1875.32
Wuhu	Anhui	31.3530	118.4180	8.0	6009.02	631.88
Wuxi	Jiangsu	31.4832	120.6517	5.00	4627.47	1621.84
Xiaogan	Hubei	30.9384	113.9883	25.00	8904.00	527.62

Table 3: Geographical and demographic characteristics of eight cities

2.2 Data Processing

Due to current limitations in mathematical modeling capabilities, it is not yet feasible to accurately fit a nonlinear relationship between two irregularly shaped areas—such as between the spatial extent of vegetation and urban and use—especially when these shapes differ significantly in geometry and distribution. As a result, a simplification strategy was adopted for the spatial selection of analysis regions. Specifically, for each city, the largest possible rectangular area (lpra) entirely contained within the city’s administrative boundary was selected. This approach allows for a more consistent and computationally manageable analysis, while still capturing a representative portion of the urban environment for comparative study.

Since the Earth can be reasonably approximated as a sphere, spherical coordinates are employed, where λ denotes longitude and ϕ denotes latitude. Under this assumption, for $\lambda \in [\alpha_1, \alpha_2]$ and $\phi \in [\beta_1, \beta_2]$, the parameterization of the Earth’s surface is given by:

$$\vec{r}(\phi, \lambda) = (a \cos \phi \cos \lambda, a \cos \phi \sin \lambda, a \sin \phi)$$

where a is the radius of the Earth.

The surface area of the region S on the sphere, corresponding to the domain D in the (θ, ϕ) coordinate system, can then be computed using the surface integral:

$$\iint_S \left| \frac{\partial \vec{r}}{\partial \phi} \times \frac{\partial \vec{r}}{\partial \lambda} \right| dS = \int_{\alpha_1}^{\alpha_2} \int_{\beta_1}^{\beta_2} a^2 \cos \phi d\phi d\lambda$$

According to the formula, a obtains 6371.393 km, and the calculation results are produced by Desmos and Excel:

City	Land Area(km^2)	lpra (km^2)	Removed Area (km^2)
Huzhou	5820	2630.92	3189.08
Jiaxing	4237	1915.88	2321.12
Shanghai	6340.5	2542.56	3797.94
Suzhou	8657.32	2373.57	6283.75
Wuhan	8569.19	3115.71	5453.48
Wuhu	6009.02	2960.85	3048.17
Wuxi	4627.47	920.83	3706.64
Xiaogan	5820	2630.92	3189.02

Table 4: Largest possible rectangular area(lpra) of eight cities



Figure 2: The selected area of eight cities

In addition, careful attention was paid to the scale factor used in the remote sensing data obtained from the MODIS (Moderate Resolution Imaging Spectroradiometer) satellite products. In the MODIS Terra Vegetation Indices (specifically MOD13C2), the Normalized Difference Vegetation Index (NDVI) values are stored as integer values that have been multiplied by a scale factor of 10,000. However, according to the MODIS data documentation, the actual scale factor required to convert these values to their true physical measurements is 0.0001. This implies that the exported raw NDVI values are numerically inflated by a factor of 100 million compared to their true scale (i.e., 10,000 divided by 0.0001). Failure to account for this discrepancy would lead to severe distortions in the interpretation of vegetation density and associated analyses.

To streamline the data extraction process and reduce computational burden, rectangular bounding boxes were defined for each selected city based on their latitude and longitude coordinates. These geographic bounding boxes were then used to extract subsets of the NDVI and LST data from the global datasets hosted on Google Earth Engine[4]. This method ensures that only the necessary spatial regions are processed and analyzed, optimizing both computational efficiency and data relevance.

2.3 Modeling Approach

To investigate the spatially varying and potentially nonlinear relationship between vegetation cover and land surface temperature (LST), this study adopts a **multivariate nonlinear regression framework** using Python. The modeling process was designed to balance computational efficiency and spatial representativeness while capturing the complexity inherent in urban ecological systems.

For each of the eight selected cities, the analysis began by identifying the largest inscribed rectangular area within the administrative urban boundary. This methodological simplification was necessitated by the current limitations in my mathematical training, which does not yet allow for modeling arbitrary, irregular polygons with nonlinear surface fitting. By constraining the study area to a maximal rectangle, the data processing steps became more tractable while still capturing a sufficiently large and representative portion of each urban landscape.

Within each selected rectangle, pixel-level values of geographic coordinates-specifically, latitude and longitude-along with NDVI were extracted from MODIS Terra products. These three variables served as the predictors in a multivariate polynomial regression model, with LST as the response variable.

$$T(\phi, \lambda, v) = \sum_{i=0}^P \sum_{j=0}^{P-i} \sum_{k=0}^{P-i-j} a_{ijk} \phi^i \lambda^j v^k \quad (1)$$

Symbol	Definition	Unit/Range
ϕ	Latitude	Degrees North (°N)
λ	Longitude	Degrees East (°E)
v	NDVI	Dimensionless, $[-1, 1]$
T	LST	Kelvin (K)
P	Polynomial Degree	Positive Integer ($1 \leq P \leq 10$)
a_{ijk}	Polynomial Coefficients	Regression-fitted constants

Table 5: Symbol definitions used in the polynomial model

Intuitively, this formula can be simply understood as:

$$\text{LST} = \mathbf{a}_0 + \mathbf{a}_1 \cdot \text{NDVI} \quad (2)$$

The choice of a polynomial model, as opposed to simpler linear models or more complex machine learning algorithms, was motivated by its balance of interpretability and flexibility in modeling curvature and interaction effects among variables. Polynomial regression allows the model to approximate a wide range of smooth functional relationships without requiring black-box methods.

To systematically explore the model complexity needed for each city, the polynomial degree was varied from 1 (linear) to 10 (highly nonlinear). For each polynomial degree, a model was trained and evaluated based on the *adjusted coefficient of determination* (R_{adj}^2), calculated using the formula:

$$R_{\text{adj}}^2 = 1 - \left(\frac{(1 - R^2)(n - 1)}{n - p - 1} \right)$$

where R^2 is the coefficient of determination, n is the number of observations, and p is the number of predictors in the model. The final model for each city was selected by identifying the degree that yielded the highest R_{adj}^2 , which indicates the best balance between goodness-of-fit and model complexity. Adjusted R^2 was chosen over the standard R^2 because it accounts for the number of predictors used, penalizing models that include irrelevant or redundant terms. This makes it a more robust criterion when comparing polynomial models of different degrees, helping to avoid overfitting and ensuring that any increase in model complexity contributes meaningfully to predictive performance.

This approach allowed the study to quantify and compare the strength of NDVI–LST relationships across cities, with each fitted model serving as a localized representation of urban vegetation’s cooling effect. Moreover, by constraining the model form to a polynomial and selecting based on statistical fit, this process ensured a replicable and scalable method that could be extended to additional cities or variables in future studies.

After obtaining a fitted polynomial function for each city, a cross-city comparative analysis was conducted on the full set of model coefficients a_{ijk} using Principal Component Analysis (PCA). PCA is a statistical technique that reduces the dimensionality of high-dimensional data while preserving as much variance as possible. In this context, PCA was applied to the collection of coefficients from all cities, treating each city as an observation and each polynomial term (e.g., $NDVI$, $NDVI^2$, $LatLon$) as a variable.

This method enabled the identification of dominant patterns and shared structures across cities, highlighting which combinations of variables most strongly contribute to the variance in model behavior. Principal components with high explained variance suggest underlying regional commonalities in how vegetation, location, and their interactions influence LST. The results of the PCA help assess whether certain polynomial terms, such as NDVI-related interactions, consistently play a key role in urban thermal dynamics, offering insights into partially generalizable mechanisms across the selected low-elevation cities within the same latitude band.

3 Data Analysis

3.1 Results from Eight Cities

This section presents the results of the multivariate nonlinear polynomial modeling applied to each of the eight selected cities. The analysis examines the strength and nature of the relationship between NDVI and LST within the predefined rectangular study areas. For each city, a range of polynomial models was systematically fitted, and the optimal model was identified based on the adjusted R^2 , which reflects the best overall explanatory power while accounting for model complexity. The resulting equations are compared across cities to assess spatial similarities and differences in the NDVI-LST relationship. Furthermore, t-tests were conducted on individual coefficients to evaluate their statistical significance, and a cross-city regression comparison was performed to identify common patterns as well as city-specific variations in the vegetation-temperature interaction.

3.1.1 Shanghai

For Shanghai, the polynomial regression model that achieved the highest adjusted coefficient of determination was a third-degree model, with an adjusted R^2_{adj} of 0.6219. This indicates that approximately 62.2% of the variance in LST can be explained by the selected combination of NDVI,

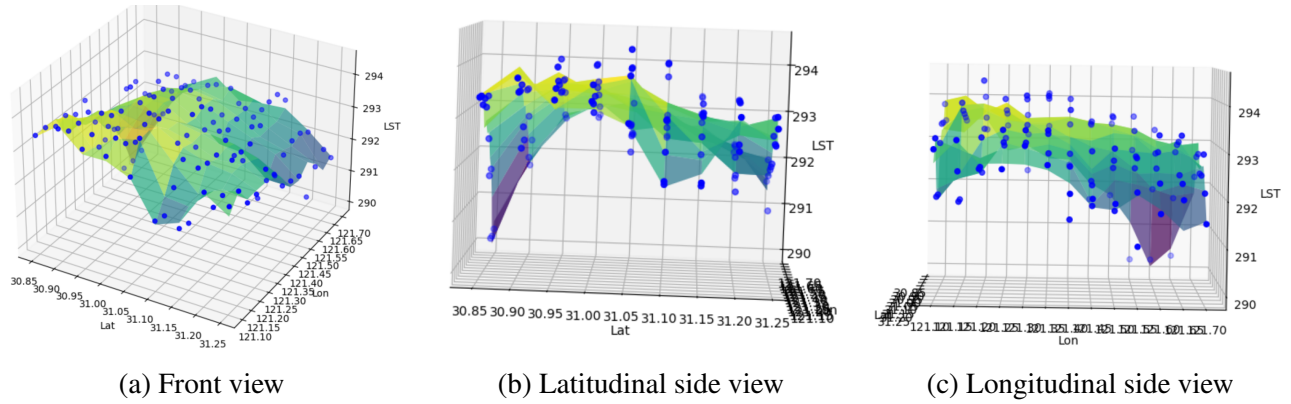


Figure 3: Three views of NDVI-LST Model of Shanghai

longitude, latitude, and their higher-order interaction terms. The final fitted equation is as follows:

$$\begin{aligned}
 \text{LST} = & 391.728890 + 128647.837104 \text{NDVI} + 100953.586406 \text{Lon} - 396393.868389 \text{Lat} \\
 & - 27504.541474 \text{NDVI}^2 + 4973.030846 \text{Lon} \cdot \text{NDVI} - 1956.270857 \text{Lon}^2 \\
 & - 27250.384131 \text{Lat} \cdot \text{NDVI} + 8748.785027 \text{Lat} \cdot \text{Lon} - 4194.741648 \text{Lat}^2 \\
 & + 412.969009 \text{NDVI}^3 + 123.384292 \text{Lon} \cdot \text{NDVI}^2 - 39.808041 \text{Lon}^2 \cdot \text{NDVI} \\
 & + 6.652120 \text{Lon}^3 + 390.462883 \text{Lat} \cdot \text{NDVI}^2 + 149.210242 \text{Lat} \cdot \text{Lon} \cdot \text{NDVI} \\
 & - 14.675540 \text{Lat} \cdot \text{Lon}^2 + 143.267676 \text{Lat}^2 \cdot \text{NDVI} - 84.113713 \text{Lat}^2 \cdot \text{Lon} \\
 & + 153.936435 \text{Lat}^3
 \end{aligned}$$

3.1.2 Suzhou (Jiangsu Province)

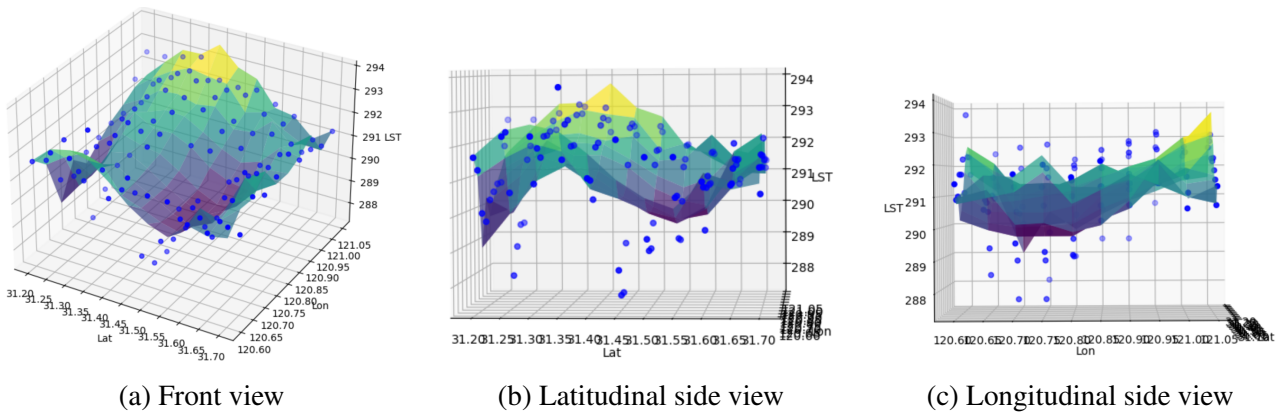


Figure 4: Three views of NDVI-LST Model of Suzhou

For Suzhou, the third-degree polynomial model provided the best fit, with an adjusted coefficient of determination of 0.3688. This suggests that approximately 36.9% of the variation in LST was explained by NDVI, geographic coordinates, and their nonlinear and interaction terms. The final fitted equation is as follows:

$$\begin{aligned} \text{LST} = & 61709.109441 - 5146316.074043 \text{NDVI} - 49646.817509 \text{Lon} + 253489.072961 \text{Lat} \\ & - 12556.405427 \text{NDVI}^2 + 77102.312153 \text{Lon} \cdot \text{NDVI} + 222.865196 \text{Lon}^2 \\ & + 31367.636206 \text{Lat} \cdot \text{NDVI} + 401.177304 \text{Lat} \cdot \text{Lon} - 9010.007744 \text{Lat}^2 \\ & + 90.185615 \text{NDVI}^3 + 38.348388 \text{Lon} \cdot \text{NDVI}^2 - 297.095366 \text{Lon}^2 \cdot \text{NDVI} \\ & - 0.996674 \text{Lon}^3 + 249.101070 \text{Lat} \cdot \text{NDVI}^2 - 169.808002 \text{Lat} \cdot \text{Lon} \cdot \text{NDVI} \\ & + 8.474057 \text{Lat} \cdot \text{Lon}^2 - 175.498654 \text{Lat}^2 \cdot \text{NDVI} - 37.979818 \text{Lat}^2 \cdot \text{Lon} \\ & + 144.821238 \text{Lat}^3 \end{aligned}$$

3.1.3 Jiaxing (Zhejiang Province)

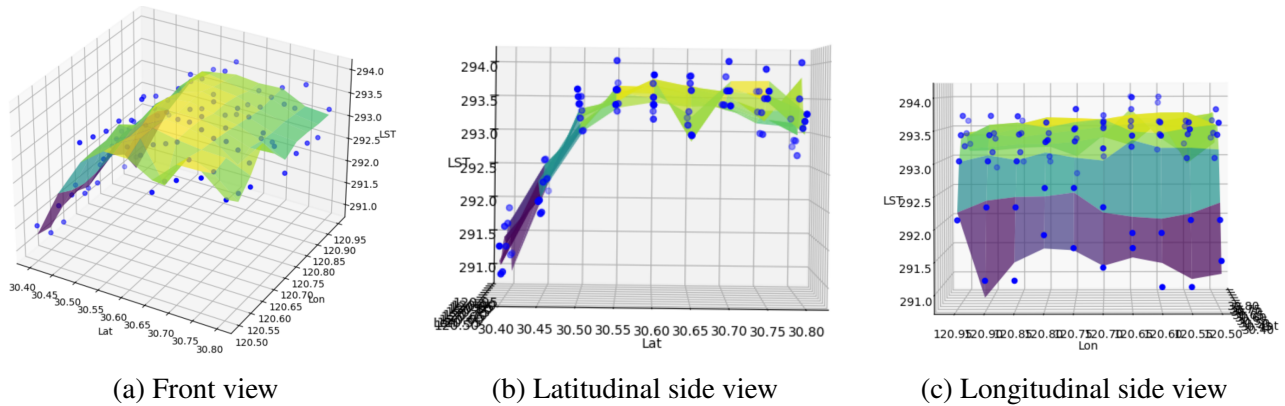


Figure 5: Three views of NDVI-LST Model of Jiaxing

For Jiaxing, the best-fitting model was a third-degree polynomial, achieving an adjusted coefficient of determination of 0.8468. This indicates that approximately 84.7% of the variance in LST was explained by the combination of NDVI, longitude, latitude, and their higher-order interaction terms, suggesting a strong and well-captured relationship between surface temperature and urban vegetation

in the region. The final fitted equation is presented below:

$$\begin{aligned} \text{LST} = & -4099.388188 + 320465.789908 \text{NDVI} - 142155.232845 \text{Lon} + 554354.808392 \text{Lat} \\ & - 17379.038276 \text{NDVI}^2 - 6328.058278 \text{Lon} \cdot \text{NDVI} + 1763.992584 \text{Lon}^2 \\ & + 4508.401459 \text{Lat} \cdot \text{NDVI} - 4514.798645 \text{Lat} \cdot \text{Lon} - 9210.079458 \text{Lat}^2 \\ & + 33.714438 \text{NDVI}^3 + 160.449121 \text{Lon} \cdot \text{NDVI}^2 + 30.547153 \text{Lon}^2 \cdot \text{NDVI} \\ & - 6.717919 \text{Lon}^3 - 67.257831 \text{Lat} \cdot \text{NDVI}^2 - 38.703394 \text{Lat} \cdot \text{Lon} \cdot \text{NDVI} \\ & + 21.339348 \text{Lat} \cdot \text{Lon}^2 + 3.563681 \text{Lat}^2 \cdot \text{NDVI} - 10.157343 \text{Lat}^2 \cdot \text{Lon} \\ & + 113.338094 \text{Lat}^3 \end{aligned}$$

Compared with other cities, the model for Jiaxing demonstrates the strongest predictive power. The surface exhibits a relatively smooth shape, with a clear decreasing trend from north to south, suggesting a clearer and more robust spatial relationship between vegetation and LST in this city.

3.1.4 Wuhu (Anhui Province)

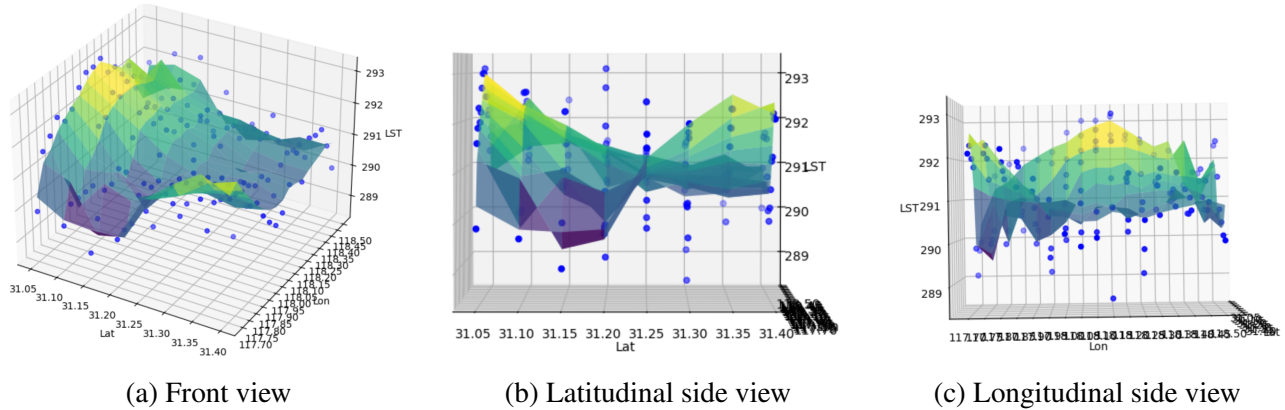


Figure 6: Three views of NDVI-LST Model of Wuhu

For Wuhu, the third-degree polynomial model was identified as the best fit, achieving an adjusted coefficient of determination of 0.5048. This indicates that approximately 50.5% of the variance in LST was explained by the model, reflecting a moderate relationship between surface temperature and the predictor variables (NDVI, longitude, latitude, and their interactions). The model captures the

nonlinear nature of these influences while avoiding overfitting.

$$\begin{aligned}
 \text{LST} = & 3265.119717 - 407641.756443 \text{NDVI} - 123416.556398 \text{Lon} + 470514.253535 \text{Lat} \\
 & - 6563.635176 \text{NDVI}^2 + 8772.909316 \text{Lon} \cdot \text{NDVI} + 2466.449869 \text{Lon}^2 \\
 & - 6865.084521 \text{Lat} \cdot \text{NDVI} - 10846.556425 \text{Lat} \cdot \text{Lon} + 5491.635095 \text{Lat}^2 \\
 & - 147.323912 \text{NDVI}^3 + 104.285871 \text{Lon} \cdot \text{NDVI}^2 - 53.848046 \text{Lon}^2 \cdot \text{NDVI} \\
 & - 11.517867 \text{Lon}^3 - 179.190743 \text{Lat} \cdot \text{NDVI}^2 + 123.398379 \text{Lat} \cdot \text{Lon} \cdot \text{NDVI} \\
 & + 52.368359 \text{Lat} \cdot \text{Lon}^2 - 121.157637 \text{Lat}^2 \cdot \text{NDVI} - 25.382096 \text{Lat}^2 \cdot \text{Lon} \\
 & - 25.891741 \text{Lat}^3
 \end{aligned}$$

3.1.5 Wuhan (Hubei Province)

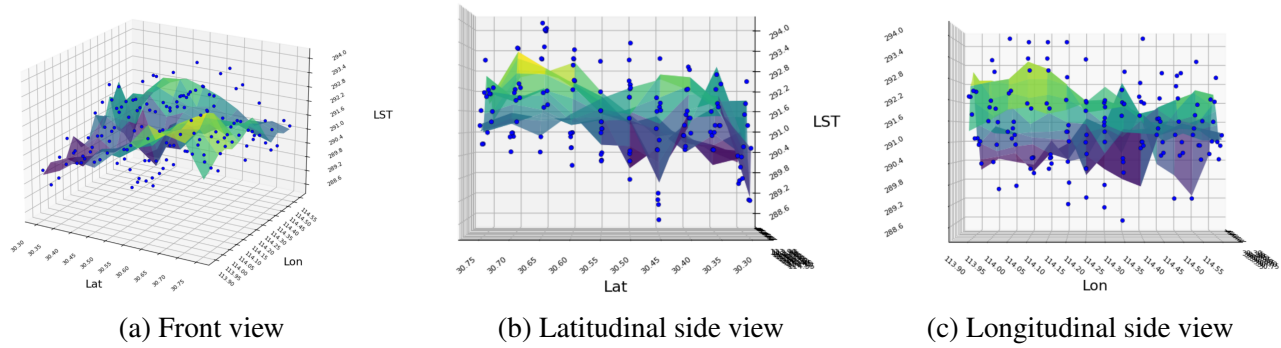


Figure 7: Three views of NDVI-LST Model of Wuhan

For Wuhan, the best-fit model was a 4th-degree polynomial, which achieved an adjusted R^2_{adj} of 0.3063. This relatively low value suggests that only about 30.6% of the variation in land surface temperature (LST) was explained by the model's predictors, indicating a weaker but still present relationship between LST and variables such as NDVI, longitude, and latitude in this region. The increased degree of the polynomial reflects the need for greater model flexibility to capture the complex interactions specific to Wuhan's urban and environmental characteristics. However, the modest R^2_{adj}

also implies the potential influence of other unmodeled factors or spatial heterogeneity within the city.

$$\begin{aligned}
 \text{LST} = & -390.579364 - 152.659251 \text{NDVI} + 6.253376 \text{Lon} - 41.093846 \text{Lat} \\
 & - 29.009818 \text{NDVI}^2 - 5812.287704 \text{Lon} \cdot \text{NDVI} + 476.886255 \text{Lon}^2 \\
 & - 1553.308040 \text{Lat} \cdot \text{NDVI} - 1498.732521 \text{Lat} \cdot \text{Lon} - 845.239586 \text{Lat}^2 \\
 & - 2.883710 \text{NDVI}^3 + 102.349265 \text{Lon} \cdot \text{NDVI}^2 + 99.563211 \text{Lon}^2 \cdot \text{NDVI} \\
 & - 2.206235 \text{Lon}^3 - 75.076262 \text{Lat} \cdot \text{NDVI}^2 + 1.079502 \text{Lat} \cdot \text{Lon} \cdot \text{NDVI} \\
 & - 41.238651 \text{Lat} \cdot \text{Lon}^2 + 122.482940 \text{Lat}^2 \cdot \text{NDVI} + 261.383212 \text{Lat}^2 \cdot \text{Lon} \\
 & - 299.545265 \text{Lat}^3 + 4823.467300 \text{NDVI}^4 + 338.018399 \text{Lon} \cdot \text{NDVI}^3 \\
 & + 13.700590 \text{Lon}^2 \cdot \text{NDVI}^2 - 11.722291 \text{Lon}^3 \cdot \text{NDVI} - 3.166150 \text{Lon}^4 \\
 & - 1426.009997 \text{Lat} \cdot \text{NDVI}^3 - 120.854765 \text{Lat} \cdot \text{Lon} \cdot \text{NDVI}^2 \\
 & + 126.770304 \text{Lat} \cdot \text{Lon}^2 \cdot \text{NDVI} + 47.709388 \text{Lat} \cdot \text{Lon}^3 \\
 & + 252.241331 \text{Lat}^2 \cdot \text{NDVI}^2 - 474.309510 \text{Lat}^2 \cdot \text{Lon} \cdot \text{NDVI} \\
 & - 267.907247 \text{Lat}^2 \cdot \text{Lon}^2 + 589.165176 \text{Lat}^3 \cdot \text{NDVI} + 666.811939 \text{Lat}^3 \cdot \text{Lon} \\
 & - 622.650540 \text{Lat}^4
 \end{aligned}$$

3.1.6 Xiaogan (Hubei Province)

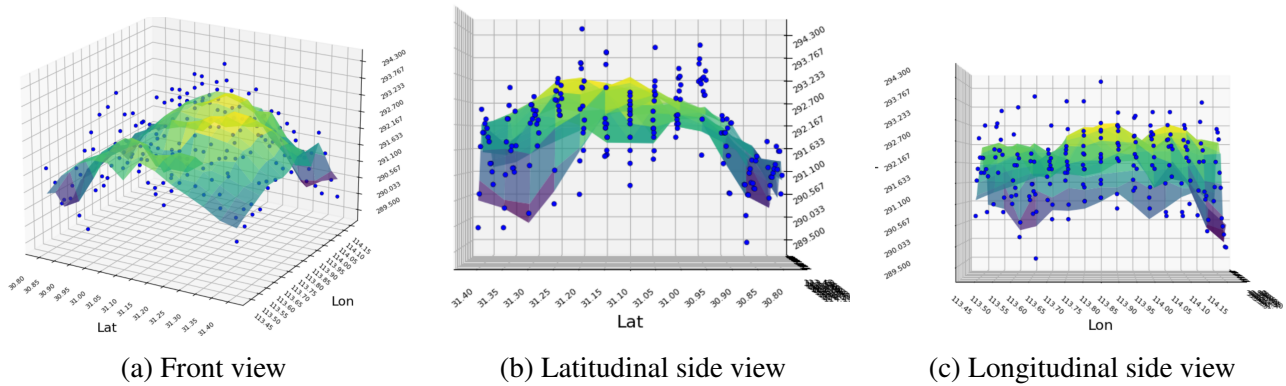


Figure 8: Three views of NDVI-LST Model of Xiaogan

For Xiaogan, the optimal polynomial degree for the regression model was 3, yielding an adjusted coefficient of determination (R_{adj}^2) of 0.5205. This indicates that about 52.1% of the variance in land

surface temperature (LST) was explained by the predictors, suggesting a moderate fit.

$$\begin{aligned} \text{LST} = & 4744.626068 - 541669.894922 \text{ NDVI} + 42498.614498 \text{ Lon} - 151129.634183 \text{ Lat} \\ & - 8277.229968 \text{ NDVI}^2 + 7730.368371 \text{ Lon} \cdot \text{NDVI} - 853.396099 \text{ Lon}^2 \\ & + 6788.791504 \text{ Lat} \cdot \text{NDVI} + 3439.489336 \text{ Lat} \cdot \text{Lon} - 1456.279693 \text{ Lat}^2 \\ & + 78.653745 \text{ NDVI}^3 + 62.915636 \text{ Lon} \cdot \text{NDVI}^2 - 22.676593 \text{ Lon}^2 \cdot \text{NDVI} \\ & + 3.780204 \text{ Lon}^3 + 33.088169 \text{ Lat} \cdot \text{NDVI}^2 - 84.355991 \text{ Lat} \cdot \text{Lon} \cdot \text{NDVI} \\ & - 13.865606 \text{ Lat} \cdot \text{Lon}^2 + 44.532645 \text{ Lat}^2 \cdot \text{NDVI} - 4.076154 \text{ Lat}^2 \cdot \text{Lon} \\ & + 20.238648 \text{ Lat}^3 \end{aligned}$$

3.1.7 Wuxi (Jiangsu Province)

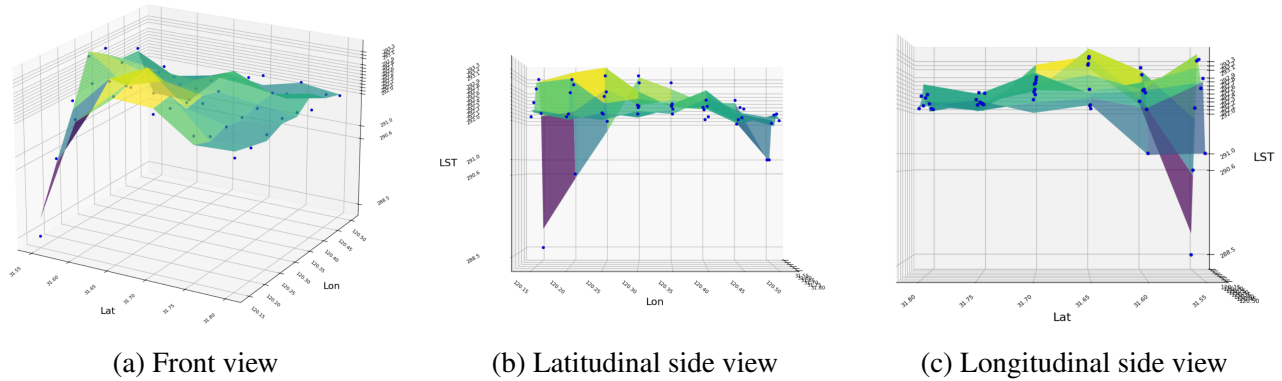


Figure 9: Three views of NDVI-LST Model of Wuxi

For Wuxi, the best-fitting model was a 3rd-degree polynomial, achieving an adjusted coefficient of determination of 0.7859, indicating a relatively strong fit where nearly 78.6% of the variance in LST is explained by the model. However, due to the land area of Wuxi being divided into two separate parts, the available sample size was quite limited. This limitation led to overfitting issues when fitting models with degrees higher than 5, where the R_{adj}^2 values exceeded 1, which is statistically invalid and a clear sign of overfitting. Therefore, restricting the model degree to 3 provided a balance between capturing complex relationships and maintaining model validity. The equation provided as follow:

$$\begin{aligned}
\text{LST} = & -22270.446943 + 1092850.927488 \text{NDVI} - 1800364.152000 \text{Lon} + 6830924.201543 \text{Lat} \\
& + 233193.441218 \text{NDVI}^2 - 41749.407451 \text{Lon} \cdot \text{NDVI} + 20523.490344 \text{Lon}^2 \\
& + 84344.571400 \text{Lat} \cdot \text{NDVI} - 41855.680053 \text{Lat} \cdot \text{Lon} - 136549.567422 \text{Lat}^2 \\
& + 1047.584166 \text{NDVI}^3 - 1992.917338 \text{Lon} \cdot \text{NDVI}^2 + 190.754369 \text{Lon}^2 \cdot \text{NDVI} \\
& - 55.912860 \text{Lon}^3 + 176.790215 \text{Lat} \cdot \text{NDVI}^2 - 86.573819 \text{Lat} \cdot \text{Lon} \cdot \text{NDVI} \\
& - 12.460895 \text{Lat} \cdot \text{Lon}^2 - 1168.591376 \text{Lat}^2 \cdot \text{NDVI} + 708.094449 \text{Lat}^2 \cdot \text{Lon} \\
& + 544.428491 \text{Lat}^3
\end{aligned}$$

3.1.8 Huzhou (Zhejiang Province)

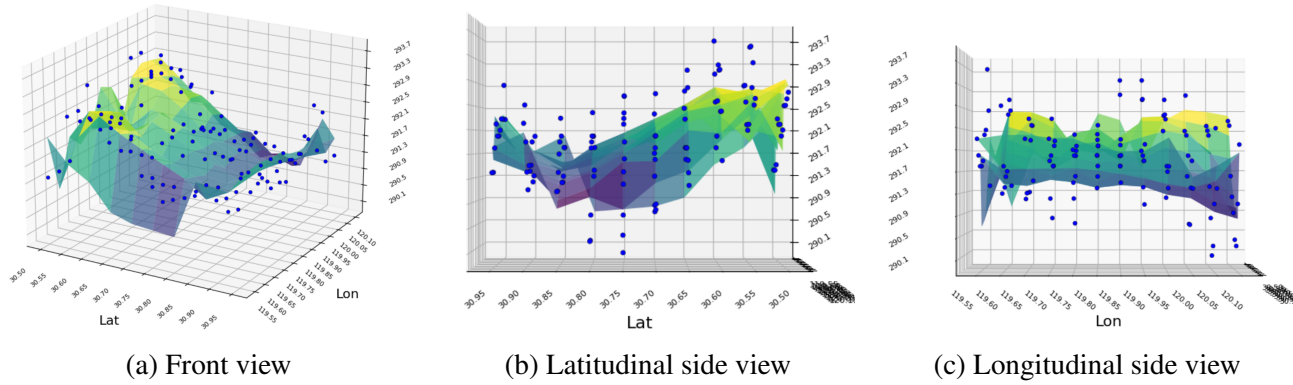


Figure 10: Three views of NDVI-LST Model of Huzhou

For Huzhou, a 3rd-degree polynomial regression model was identified as the best fit, explaining approximately 48.9% of the variability in land surface temperature (LST), as indicated by an adjusted R^2 of 0.4889.

$$\begin{aligned}
\text{LST} = & -5344.141271 - 413585.594116 \text{NDVI} - 62209.090221 \text{Lon} + 248404.900020 \text{Lat} \\
& - 2903.515120 \text{NDVI}^2 + 3570.878194 \text{Lon} \cdot \text{NDVI} + 451.894749 \text{Lon}^2 \\
& + 13079.355238 \text{Lat} \cdot \text{NDVI} + 494.835721 \text{Lat} \cdot \text{Lon} - 9162.306923 \text{Lat}^2 \\
& - 90.020396 \text{NDVI}^3 + 20.034278 \text{Lon} \cdot \text{NDVI}^2 - 9.413802 \text{Lon}^2 \cdot \text{NDVI} \\
& - 0.819667 \text{Lon}^3 + 20.348286 \text{Lat} \cdot \text{NDVI}^2 - 43.341540 \text{Lat} \cdot \text{Lon} \cdot \text{NDVI} \\
& - 5.070566 \text{Lat} \cdot \text{Lon}^2 - 128.680957 \text{Lat}^2 \cdot \text{NDVI} + 12.025854 \text{Lat}^2 \cdot \text{Lon} \\
& + 84.594600 \text{Lat}^3
\end{aligned}$$

3.2 Quantitative Comparison among Cities

Across the eight cities analyzed, both the adjusted R^2_{adj} values and best-fitting polynomial degrees varied, reflecting underlying differences in how effectively NDVI and spatial variables (latitude and longitude) explain variations in land surface temperature (LST). These differences provide insight into each city's land-use composition, vegetation cover, and urban structure.

Jiaxing stood out with the highest adjusted R^2 of 0.8468, suggesting a very strong relationship between LST and the predictor variables. This high explanatory power may indicate a relatively uniform urban layout and consistent vegetation patterns, which the model captures well using a degree 3 polynomial. Likewise, Wuxi and Shanghai achieved high adjusted R^2 values of 0.6935 and 0.6219, respectively. These results point to effective model performance, though in the case of Wuxi, the use of a degree 4 polynomial and the observation that higher-degree models produced implausibly high R^2_{adj} values (greater than 1) suggest overfitting, likely due to the limited sample coverage caused by fragmented land boundaries.

On the other end of the spectrum, Wuhan (0.3063) and Suzhou (0.3688) exhibited relatively low adjusted R^2 values, indicating weaker model performance. This may reflect greater heterogeneity in land cover types, uneven distribution of green space, or other unmodeled influences like elevation or impervious surfaces. For Wuhan in particular, the requirement for a degree 4 polynomial despite the low R^2_{adj} suggests that the LST patterns are complex but not well-explained by NDVI and spatial coordinates alone.

In terms of polynomial complexity, most cities reached optimal model performance with a degree 3 polynomial, indicating that moderate nonlinear interactions between NDVI, latitude, and longitude sufficiently captured the trends. However, Wuhan required degree 4. From the model perspective, the fluctuations in Wuhan are rather complex, and it is not possible to visually identify the patterns, reflecting either more complex temperature-vegetation relationships or possibly a need to account for subtle interactions not captured by simpler models.

Overall, these results suggest that urban form and vegetation-temperature relationships vary significantly across cities. Cities with compact or well-vegetated cores (like Jiaxing) may exhibit more predictable cooling patterns due to vegetation, while more sprawling or heterogeneous cities (like Wuhan) show more diffuse or inconsistent trends. These insights highlight the value of using city-specific modeling approaches when analyzing environmental patterns, rather than assuming a one-size-fits-all model.

3.3 Coefficient Patterns Analysis

Given that PCA requires a horizontal comparison of coefficients under identical polynomial terms, the model for Wuhan needs to be refitted using a third-degree polynomial to ensure consistency with the other cities.

$$\begin{aligned} \text{LST} = & -49412.450451 - 1870321.546042 \text{NDVI} + 84104.511126 \text{Lon} - 292709.274605 \text{Lat} \\ & + 42226.076466 \text{NDVI}^2 + 31148.168315 \text{Lon} \cdot \text{NDVI} - 1333.215941 \text{Lon}^2 \\ & + 5144.150294 \text{Lat} \cdot \text{NDVI} + 4159.288346 \text{Lat} \cdot \text{Lon} + 1826.441778 \text{Lat}^2 \\ & + 66.961608 \text{NDVI}^3 - 444.803454 \text{Lon} \cdot \text{NDVI}^2 - 128.451836 \text{Lon}^2 \cdot \text{NDVI} \\ & + 5.068105 \text{Lon}^3 + 279.310623 \text{Lat} \cdot \text{NDVI}^2 - 49.670939 \text{Lat} \cdot \text{Lon} \cdot \text{NDVI} \\ & - 11.889602 \text{Lat} \cdot \text{Lon}^2 + 4.897422 \text{Lat}^2 \cdot \text{NDVI} - 23.615603 \text{Lat}^2 \cdot \text{Lon} \\ & + 9.289480 \text{Lat}^3 \end{aligned}$$

Based on the reconstructed third-degree polynomial, the adjusted coefficient of determination is $R_{\text{adj}}^2 = 0.2857$, which is only 2% lower than the best-fit fourth-degree model for Wuhan ($R_{\text{adj}}^2 = 0.3063$). Such a minor difference is considered acceptable for the purposes of cross-city comparison.

Principal Component Analysis (PCA) was employed to reduce the dimensionality and visualize the structure of the multivariate polynomial regression coefficients that model the urban heat island (UHI) effect across eight cities: Shanghai, Suzhou, Jiaxing, Wuhu, Wuhan, Xiaogan, Wuxi, and Huzhou. In the original dataset, each city was characterized by a set of 20 polynomial coefficients, including the constant term, NDVI, geographic coordinates (longitude and latitude), and their higher-order interaction terms (Constant, NDVI, Lon, Lat, NDVI^2 , $\text{Lon} \cdot \text{NDVI}$, Lon^2 , $\text{Lat} \cdot \text{NDVI}$, $\text{Lat} \cdot \text{Lon}$, Lat^2 , NDVI^3 , $\text{Lon} \cdot \text{NDVI}^2$, $\text{Lon}^2 \cdot \text{NDVI}$, Lon^3 , $\text{Lat} \cdot \text{NDVI}^2$, $\text{Lat} \cdot \text{Lon} \cdot \text{NDVI}$, $\text{Lat} \cdot \text{Lon}^2$, $\text{Lat}^2 \cdot \text{NDVI}$, $\text{Lat}^2 \cdot \text{Lon}$, Lat^3). These coefficients capture the complex and nonlinear relationships between land surface temperature (LST), vegetation index (NDVI), and spatial position.

PCA, an unsupervised dimensionality reduction technique, was applied to the standardized coefficient matrix. By performing eigendecomposition on the covariance matrix, the original 20-dimensional space was transformed into a lower-dimensional space defined by the first two principal components (PC1 and PC2). Each principal component is a linear combination of the original variables, ordered by the amount of variance it explains in the data. The PCA scatter plot displays each city as a point projected into the new coordinate system, where the relative distances between points represent the similarity of their polynomial coefficient structures: the closer the points, the more similar the underlying regression structure.

According to the results, PC1 accounts for 66.4% of the total variance and captures the dominant pattern of variation among cities. Along this axis, Wuxi is clearly separated from the other six cities, suggesting a distinct coefficient structure, potentially due to its unique environmental or

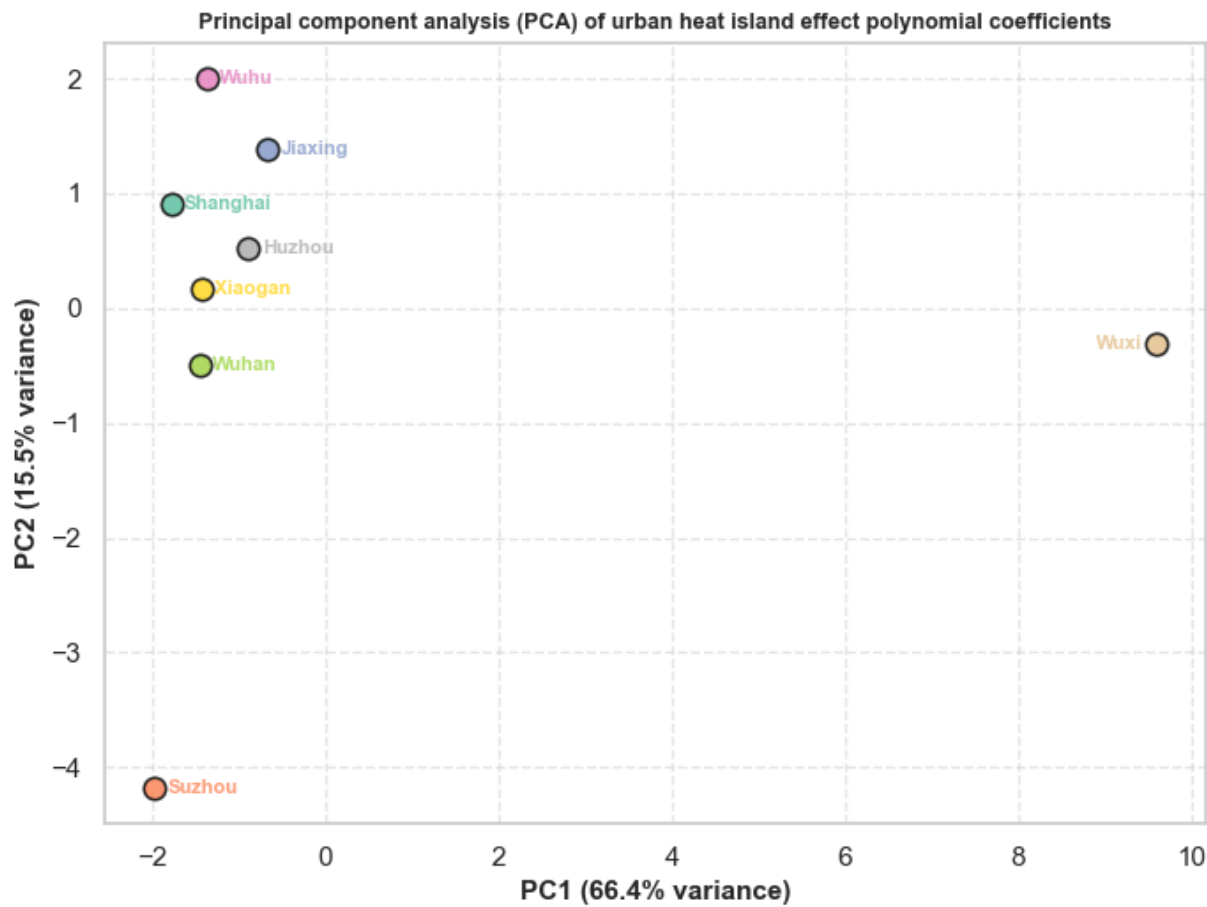


Figure 11: PCA of the coefficients among eight cities

urban characteristics. PC2 explains 15.5% of the variance and is often associated with subtler or higher-dimensional variation. Suzhou is notably distant from the rest along PC2, indicating another dimension of differentiation in its regression profile. In contrast, the five cities clustered in the upper-right quadrant-Shanghai, Jiaxing, Huzhou, Wuhu, and Xiaogan-exhibit similar polynomial structures, implying shared underlying patterns in the way vegetation and spatial variables interact to influence surface temperature.

3.4 Geographical Patterns Analysis

To assess the extent to which the fitted model reflects the underlying regularity between NDVI and LST from a geographical perspective, a comparative analysis is conducted. Specifically, a three-dimensional visualization of real-world topographic variations and vegetation distribution is juxtaposed with the modeled surface generated through multivariate regression. This comparison aims to identify interpretable spatial patterns and transitional features that might bridge the empirical terrain-vegetation structure with the model's mathematical representation. By integrating physical geography with statistical modeling, this section explores whether topography-mediated vegetation distribution can help explain the heterogeneity in temperature responses captured by the fitted surface.

3D Terrain and Vegetation (NDVI) of Shanghai

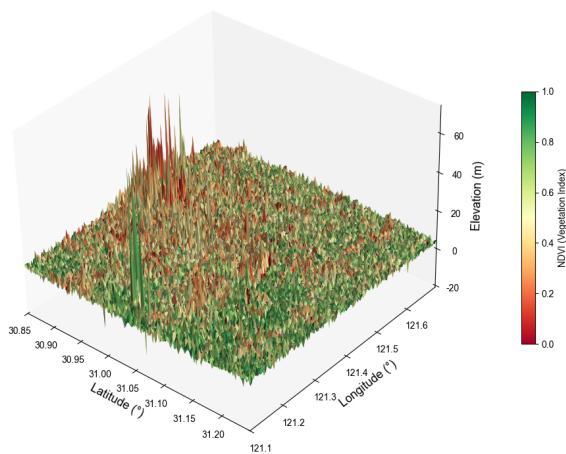


Figure 12: Real 3D terrain with vegetation distribution of Shanghai

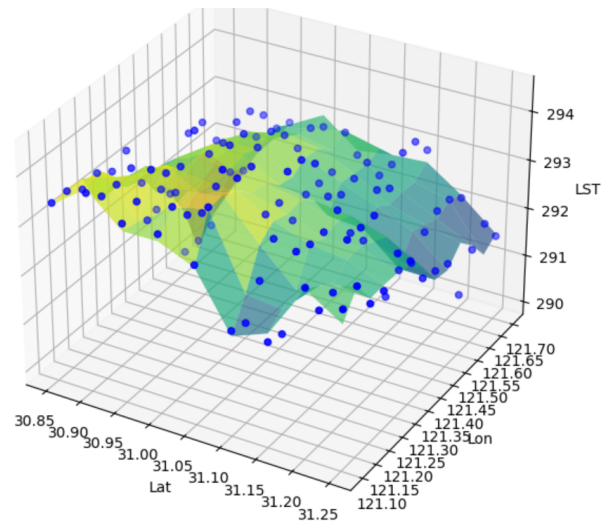


Figure 13: NDVI-LST model of Shanghai

Shanghai: Overall, the vegetation distribution in Shanghai is highly scattered. However, there is a general trend of increasing NDVI with latitude, which is consistent with the pattern shown by the model. However, according to the model, the predicted LST reaches its peak around the coordinates (30.85, 121.15), while in reality, the vegetation in this area is not particularly sparse. Instead, vegetation is more limited around the city center (30.85, 121.4).

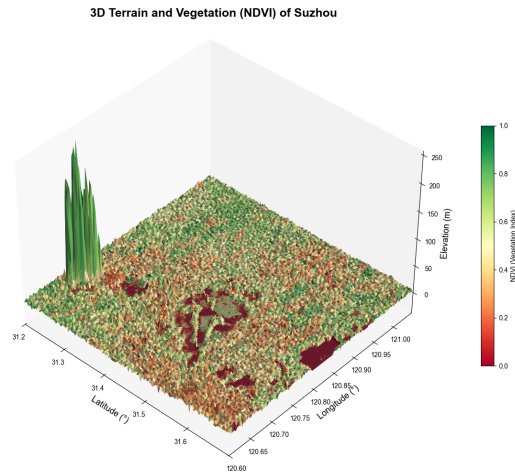


Figure 14: Real 3D terrain with vegetation distribution of Suzhou

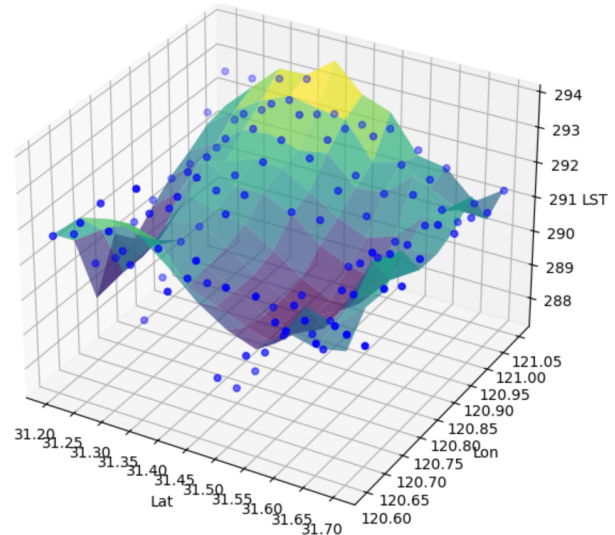


Figure 15: NDVI-LST model of Suzhou

Suzhou: The significant fluctuations in Suzhou's model can be explained by its topography. The red, flat depressions on the 3D topographic map correspond to Suzhou's lakes, and the large bodies of water effectively contribute to cooling the surrounding environment. The depressions shown in the model also reflect this feature well. In addition, there is a prominent elevation around the coordinates (31.3, 120.65) on the topographic map, where the altitude can reach up to 200 meters, and the vegetation density is high. Correspondingly, the model shows a noticeable temperature drop in this area.

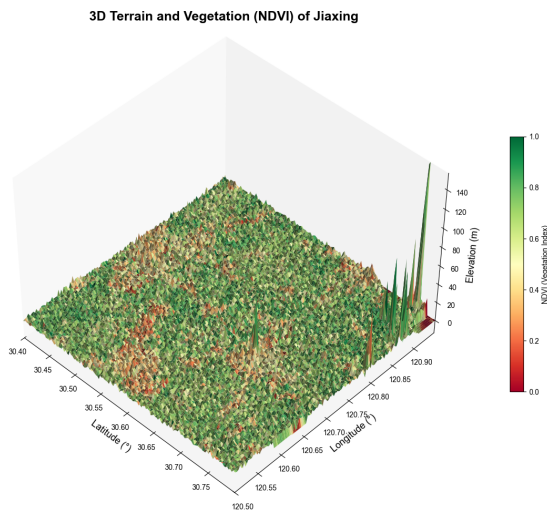


Figure 16: Real 3D terrain with vegetation distribution of Jiaxing

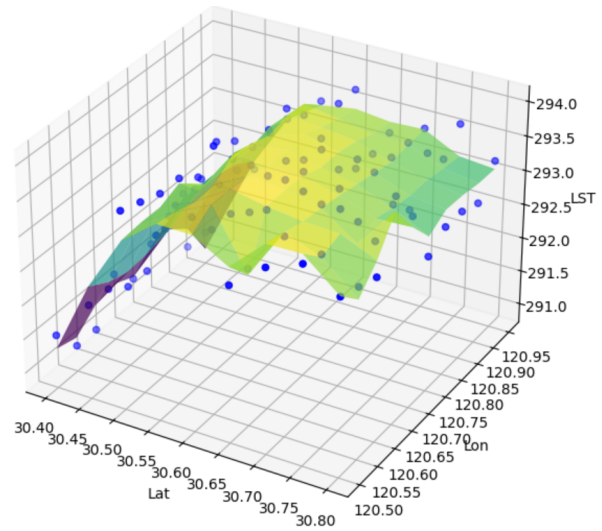


Figure 17: NDVI-LST model of Jiaxing

Jiaxing: In terms of topography and vegetation distribution alone, Jiaxing features a very flat terrain, with only a small mountainous area in the upper right corner. Vegetation is generally dense

throughout the region. On the other hand, the model reveals a clear upward trend in LST corresponding to changes in latitude. Although the variation occurs at a decimal level, the overall pattern is significant.

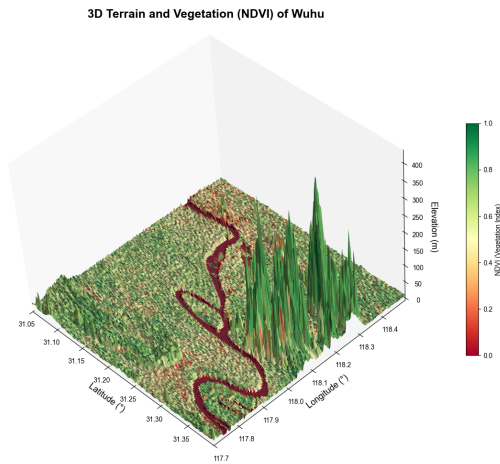


Figure 18: Real 3D terrain with vegetation distribution of Wuhu

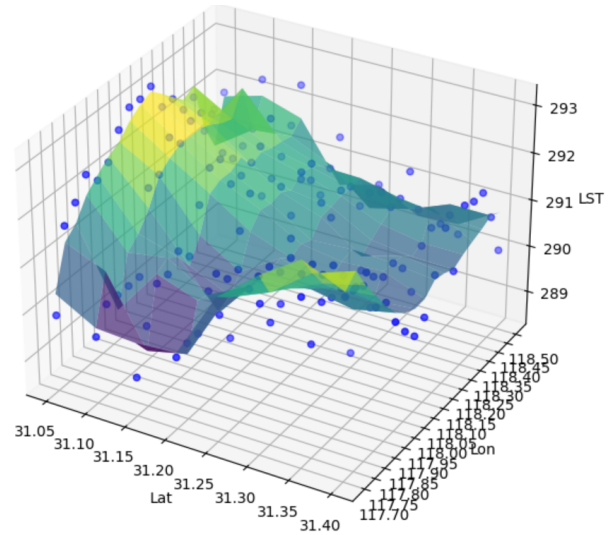


Figure 19: NDVI-LST model of Wuhu

Wuhu: Wuhu's topography is particularly distinctive, as the city is divided by the Yangtze River, with mountains on both sides exhibiting significant elevation differences. The highest elevation on the left side of the river is around 50 meters, while on the right side it reaches up to 300 meters. Vegetation is sparser near the riverbanks and denser near the surrounding mountains. In the model, areas closer to the mountains show significantly lower LST, whereas the plains and regions along the Yangtze River exhibit much higher temperatures, forming a bimodal distribution.

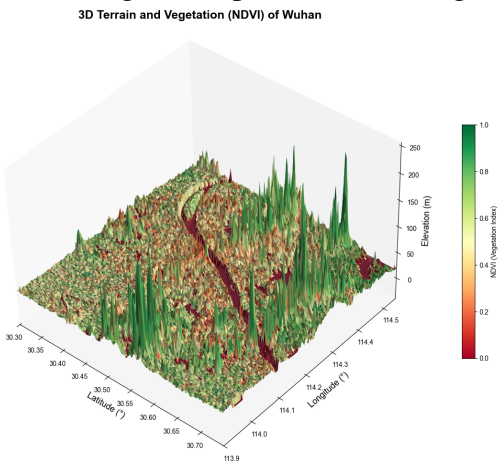


Figure 20: Real 3D terrain with vegetation distribution of Wuhan

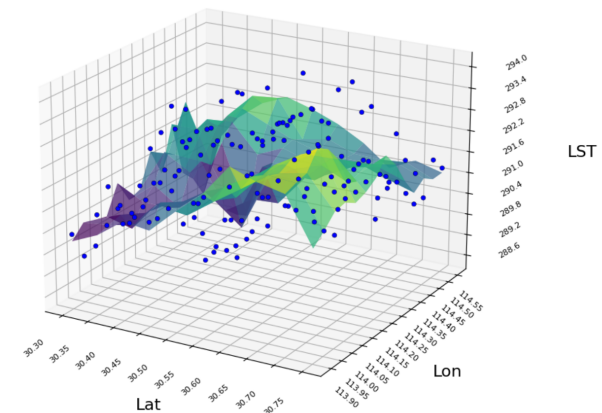


Figure 21: NDVI-LST model of Wuhan

Wuhan: Compared to other cities, Wuhan has relatively sparse vegetation overall, and its topography is also distinct, characterized by scattered mountain clusters. Except for the mountainous areas, vegetation is scarce along the Yangtze River and in the relatively flat regions. As a result, the fitted model for Wuhan performs the worst among all cities; due to the complexity of the terrain, even the best-fitting model can only explain 30% of the pattern.

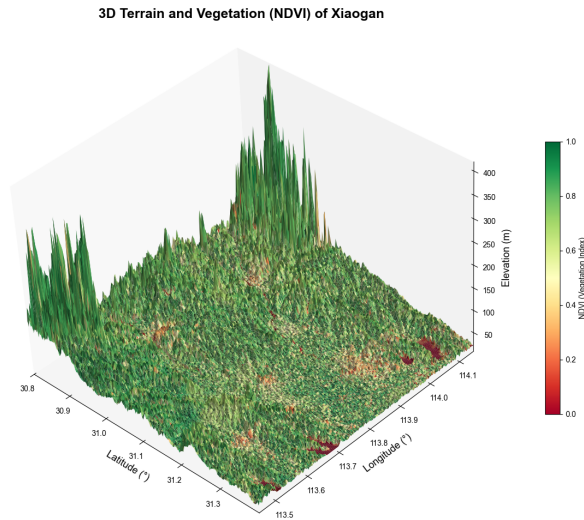


Figure 22: Real 3D terrain with vegetation distribution of Xiaogan

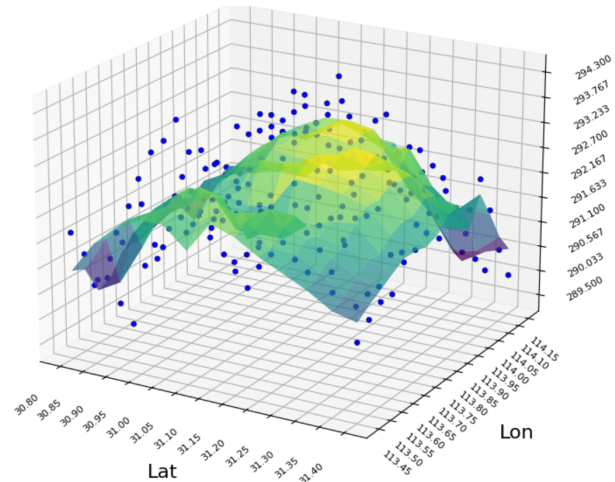


Figure 23: NDVI-LST model of Xiaogan

Xiaogan: As the boundary of China's third topographic step, Xiaogan is situated near mountainous areas and exhibits noticeable elevation gradients. The vegetation is dense, and although temperatures slightly decrease near the mountains, the overall temperature fluctuation across the city remains within approximately one Kelvin.

Wuxi: The overall terrain of Wuxi is relatively flat, except for a small hill located at (31.8, 120.2) with an elevation of approximately 200 meters. The vegetation index on the hill is favorable, but there is a large area of vegetation scarcity at its base. From the model results, a clearly noticeable outlier affects the overall fitting performance. Aside from this, the LST shows minimal fluctuation, with a peak observed at (31.65, 120.2).

Huzhou: Similar to Xiaogan, Huzhou has many mountains, with abundant vegetation on the hills and relatively scarce vegetation in the plains. Since most of the mountains are located in the higher latitudes, the model displays a clear gradient, with significantly lower temperatures in the mountainous areas compared to the plains. However, although part of the topography is obscured by mountains, the area around the coordinates (30.8, 120.1) is also a plain with relatively sparse vegetation, yet the model shows a significant drop in temperature.

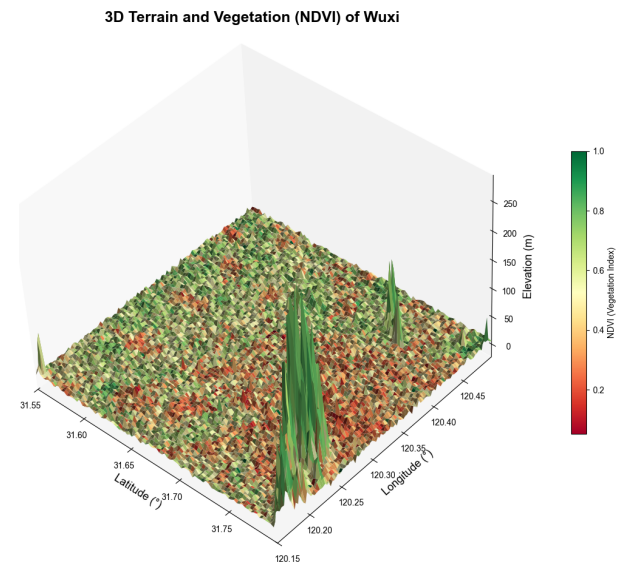


Figure 24: Real 3D terrain with vegetation distribution of Wuxi

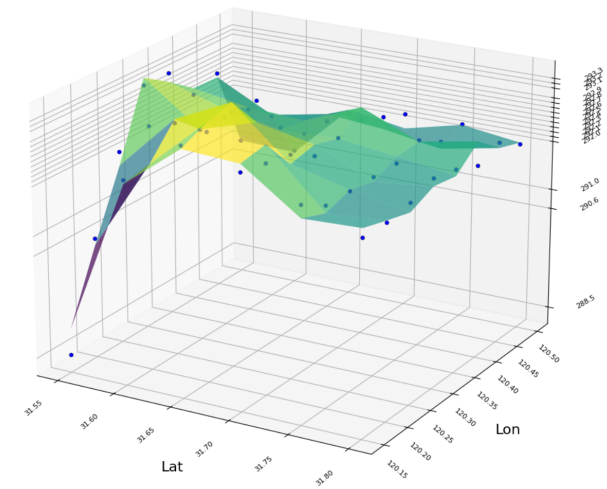


Figure 25: NDVI-LST model of Wuxi

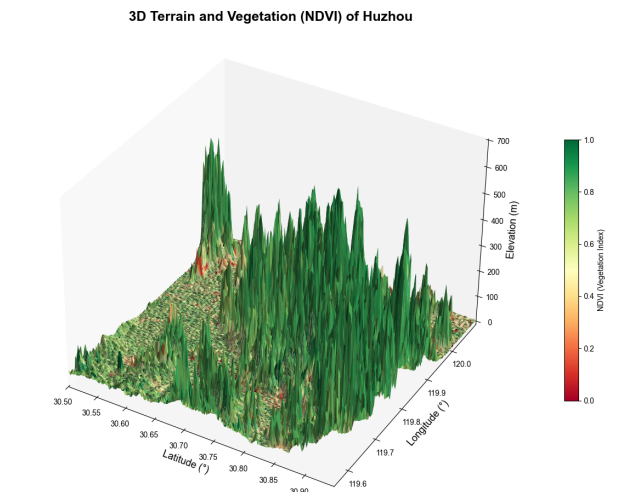


Figure 26: Real 3D terrain with vegetation distribution of Huzhou

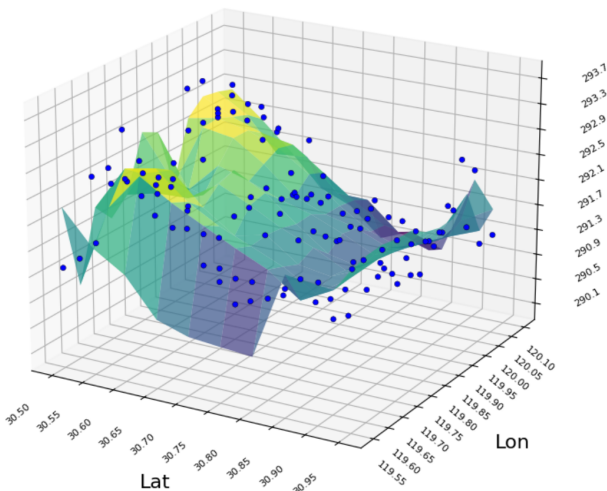


Figure 27: NDVI-LST model of Huzhou

4 Discussion

4.1 Summary of Findings

4.1.1 Regression Model

The NDVI-LST regression model revealed that vegetation index and geographical factors (latitude, longitude, and their interactions) significantly affect land surface temperature (LST). While most cities showed expected patterns where higher vegetation correlated with lower temperatures, outliers due to local terrain or land use disrupted the model's fit. For instance, an anomalously high LST was found at the foot of a single hill in Wuxi despite good vegetation on the slope, skewing the overall trend. In contrast, cities like Suzhou and Changzhou displayed consistent vegetation-temperature relationships without extreme deviation.

4.1.2 Principal Component Analysis (PCA)

PCA identified the dominant factors influencing LST variation across cities. In urbanized areas, anthropogenic heat and sparse vegetation drove higher LST values, while mountainous or forested zones exhibited lower LST due to dense vegetation cover. The dimensionality reduction confirmed that vegetation and elevation jointly explained most of the variance in temperature distribution.

4.1.3 Real 3D Terrain with Vegetation Mapping

By integrating NDVI data with 3D elevation models, a clearer spatial correlation between vegetation distribution, topography, and LST emerged. Cities like Xiaogan and Huzhou, characterized by mountainous regions, showed a strong temperature gradient-cooler temperatures prevailed in higher elevation zones with rich vegetation. Notably, exceptions were found where certain low-lying plains unexpectedly displayed lower LST, potentially due to microclimatic influences not captured by elevation or NDVI alone.

4.2 Various Driving Factors Influencing LST

It is evident that the urban environment is a complex and dynamic system shaped by the interplay of numerous factors; it can never be fully captured by a single variable. Based on the models fitted through our analytical methods, the average adjusted coefficient of determination R^2 value reached 0.5555. This indicates that, under comparable geographic conditions, the Normalized Difference Vegetation Index (NDVI) alone accounts for 55.55% of the variation in Land Surface Temperature (LST), strongly suggesting a robust correlation between the two.

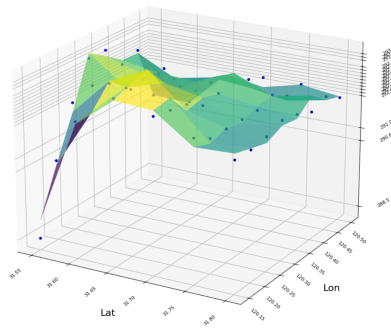
Moreover, as shown in the results of Principal Component Analysis (Figure 11), the first principal component (PC1) explains 66.4% of the total variance in the polynomial characteristics across the

cities. Except for Wuxi, all cities are closely aligned in this dimension. The inclusion of PC2, which explains an additional 15.5% of the variance, reveals a distinguishable separation between Suzhou and the remaining five cities. This degree of clustering provides compelling evidence of shared NDVI-LST patterns among the five cities, thereby supporting our initial hypothesis: there exists a consistent and generalizable relationship between NDVI and LST.

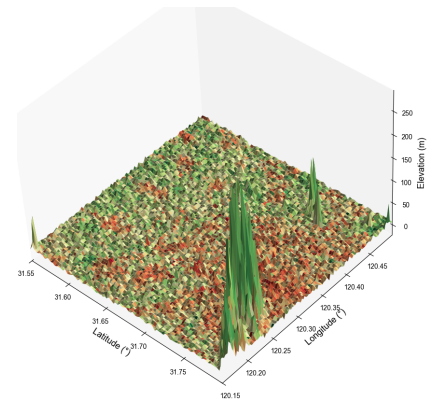
Based on the results of PCA, the NDVI-LST fitting polynomials of the eight cities can be clustered into three groups: Wuxi, Suzhou, and the remaining five cities. However, it must also be acknowledged that such patterns are only observed consistently across the five cities. The deviations exhibited by Suzhou and Wuxi suggest the presence of unique local factors, which warrant further investigation with a larger sample size.



(a) Map with study area



(b) NDVI-LST model



(c) 3D terrain

Figure 28: Three figures of Wuxi

The primary issue in Wuxi arises from limitations in mathematical capability. Currently, only regular rectangular regions can be used for nonlinear regression. However, Wuxi's land area is split into two separate parts (Figure 28a), resulting in the largest extractable rectangle covering only 19.8% of the total area. Moreover, due to this constraint, the final rectangle includes only 48 data points (8×6), meaning that each point has a significant impact on the overall fitting result. Figure 31b shows the fitting outcome without any data preprocessing, where a single extremely low value noticeably drags down the entire surface. However, after attempting to remove that point, the fitting performance deteriorated further, indicating an actual downward trend in temperature at that location.

The red star marked on the map indicates the outlier identified by the model. This location corresponds to a densely vegetated park situated near the sea, which reasonably explains its lower land surface temperature. Notably, according to Figure 30, given that Wuxi's topography remains unchanged, the lower section of the map represents the urban area of Wuxi, near the city center, where

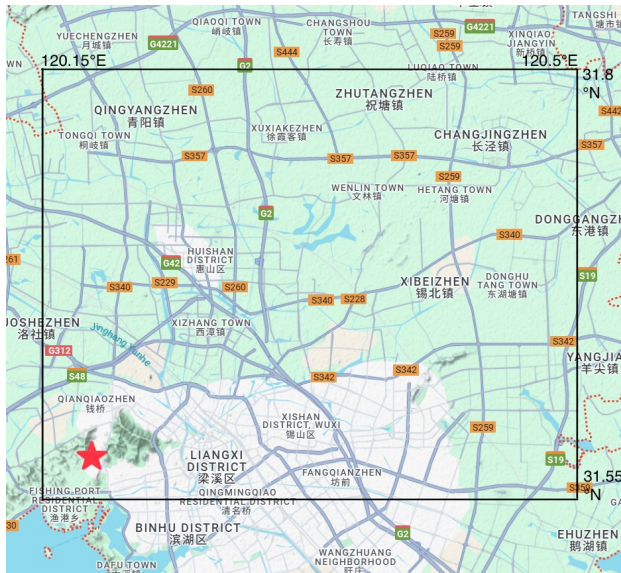


Figure 29: Map of the study area of Wuxi

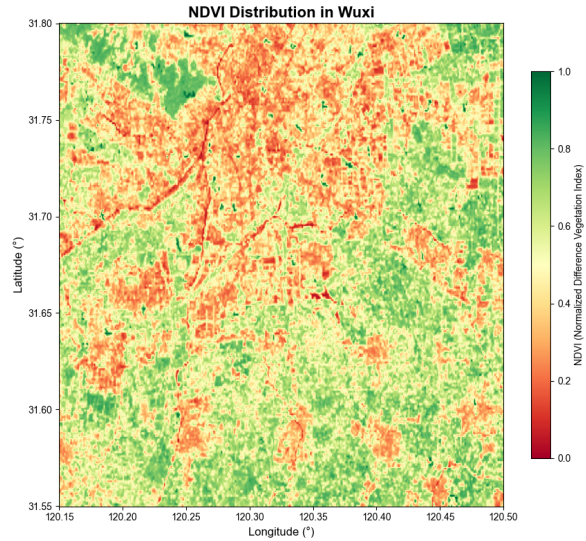
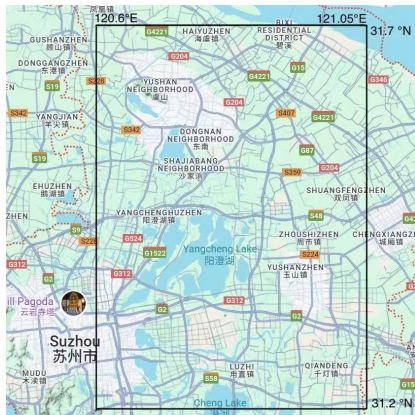
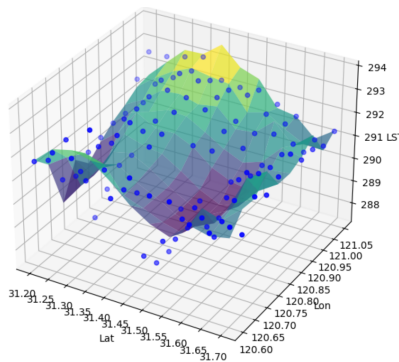


Figure 30: NDVI distribution of Wuxi

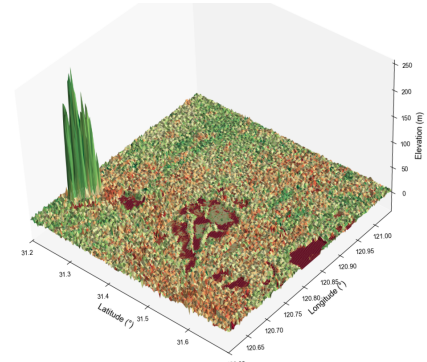
population density is higher. However, its NDVI is significantly higher than that of suburban and rural areas. A plausible explanation is that the suburbs are primarily occupied by industrial zones, which leads to a decline in vegetation maintenance. Nonetheless, although vegetation in the urban area appears abundant, the LST within the urban range still reaches the peak value of the entire model, demonstrating the significant impact of urbanization on land surface temperature.



(a) Map with study area



(b) NDVI-LST model



(c) 3D terrain

Figure 31: Three figures of Suzhou

4.3 Limitations

5 Conclusion

6 Acknowledgment

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Nomenclature

h	Planck constant
GEE	Google Earth Engine
LA	Land Area
lpra	Largest Possible Rectangular Area
LST	Land Surface Temperature
NBS	National Bureau of Statistics
NDVI	Normalized Difference Vegetation Index
PCA	Principal Components Analysis
PD	Population Density
PUR	Permanent Urban Residents
UHI	Urban Heat Island

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7 Appendix A